CYB 203: CYBERCRIME, LAW AND COUNTERMEASURES

COURSE CONTENT

The course covers:

- General introduction to cybercrime.
- Definition of cybercrime.
- Types and categories of cybercrime and threats to the national critical infrastructure.
- Investigation process and procedure for cybercrime. Strategies of cybercrime perpetrators.
- Possible ways of curbing/preventing them.
- Technical aspects of computer cybercrime investigation.
- Concepts, trends, and methods in computer and network forensics investigations.
- Skill and knowledge in digital evidence collection and evaluation.
- Policies, legal issues, international jurisdiction, and privacy issues.
- Introduction to cyber law and countermeasures.
- Cyber law application at the international and national levels with examples from European, North American, South American, and Asian countries.
- The cyber law framework in Nigeria.
- Challenges and opportunities for cyber law and countermeasure enforcement in Nigeria.

LESSON1: Understanding the crime
Modules 1 to 5: cybercrime, types, perpetrators, social engineering and critical infrastructure.

LESSON2: Investigating the crime
Modules 6 to 9: investigation, computer forensics, network forensics, digital evidence and chain of custody.

LESSON3: Understanding the law
Modules 10 to 15: policies, privacy, jurisdiction, international cyber law and Nigerian cyber law.

LESSON4: Preventing the crime
Modules 16 to 20: countermeasures, university security, ethics, Nigerian challenges and opportunities.

Relationship between **crime → investigation → law → prevention**, rather than treating the course as a collection of unrelated topics.

LESSON 1: UNDERSTANDING THE CRIME

1 Introduction to cybercrime

Computers, mobile devices, and the Internet have transformed how people communicate, learn, conduct business, store information, and access government services.

However, the same technologies that create opportunities can also be abused.

A person can use a computer to steal information, deceive another person, damage computer systems, distribute malicious software, commit financial fraud or interfere with important services.

These unlawful activities are generally described as **cybercrime**.

Cybercrime is therefore not simply "crime on the Internet." It covers a broad range of criminal activities involving computers, networks, digital devices, information systems or electronic data.

2 Why should students study cybercrime?

A Computer Science graduate may eventually work as:

• A software developer
• Network administrator
• System administrator
• Database administrator
• Cybersecurity analyst
• Digital forensic examiner
• IT auditor
• Security consultant
• ICT manager
• Incident response analyst
• Academic or researcher

In each of these professions, the individual may encounter cybercrime.

For example, a system administrator may discover that an employee's account has been used to access confidential information. A cybersecurity analyst may detect suspicious network

traffic. A digital forensic examiner may be asked to examine a laptop involved in a criminal investigation.

Therefore, students need both **technical knowledge and legal awareness**.

3 Cybercrime as a technological and social problem

Cybercrime should not be viewed only as a technical problem.

It involves:

Technology + People + Money + Law + Society

For example, phishing involves technology, but the attack's success may depend on human trust.

A criminal may create a convincing message claiming:

"Your university account will be disabled unless you verify your account."

The victim may click the link and unknowingly provide information.

The technology facilitates the crime, but human behaviour contributes to its success.

4 Basic characteristics of cybercrime

Cybercrime often has the following characteristics:

1. Borderless nature

The offender may be in one country, the victim in another country, and the relevant server in a third country.

2. Speed

Digital information can be copied and transmitted extremely quickly.

3. Anonymity

Criminals may attempt to hide their identities using fake accounts, compromised systems or other techniques.

4. Scalability

One malicious message can potentially reach thousands of people.

5. Dependence on evidence

Digital evidence can be fragile and can change quickly.

6. Technical complexity

Investigators may need specialist knowledge of computers, networks, cloud systems, databases and mobile devices.

CLASSROOM DISCUSSION

Ask students:

Question: Is cybercrime a new type of crime or an old type of crime using new technology?

Expected discussion:

Many crimes existed before computers. Fraud, theft, impersonation, and extortion, for example, are not new. Technology has changed how these crimes can be committed, increased their speed, and expanded their geographical reach.

LESSON 2: DEFINITION AND CONCEPT OF CYBERCRIME

1 Definition

Cybercrime can be defined generally as:

Any unlawful activity in which a computer, computer system, network, digital device or digital information is used as a tool, target, environment or source of evidence.

A computer is just the physical electronic device, while a computer system is the complete combination of that hardware, software, and users working together to perform tasks

This definition is useful because computers can play different roles in crime.

2 Computer as the target

Here, the computer system itself is attacked.

Examples include:

• Unauthorised access
• Malware infection
• Data destruction
• Denial of service
• System sabotage

Example

An attacker gains unauthorised access to a university server and deletes student records.

The server is the target.

3 Computer as the tool

Here, the computer is used to commit another crime.

Examples:

Definitions of Cyber Terms

- **Online fraud**: A crime where a person lies or hides facts online to get money or data from you, or uses any deceptive scheme using the Internet to trick victims out of money, property, or sensitive data.
- **Identity theft**: Stealing someone's private personal information, like a name or account number, to pretend to be them for money or illegal acts.
- **Phishing**: Sending fake emails or messages that look real to trick people into giving away passwords or bank details.
- **Cyberstalking**: Using the internet, email, or social media to harass, track, or scare a specific person repeatedly.
- **Online impersonation**: Pretending to be another real person online—often by creating fake social media profiles—to trick others or damage that person's reputation.

Example

A criminal uses email to impersonate a university bursar and deceive an employee into sending money to a fraudulent account.

The computer is being used as a tool.

4 Computer as evidence

Sometimes the computer itself is not the victim.

Instead, it contains evidence.

For example:

A suspect may use a computer to communicate with another person. Investigators may later recover relevant emails, documents, messages, or transaction records.

The computer becomes a source of evidence.

5 Categories of cybercrime

Cybercrime can be classified in several ways.

1: Crimes against individuals

Examples:

• Identity theft
• Cyberstalking
• Online harassment
• Phishing
• Account compromise
• Online impersonation

2: Crimes against organisations

Examples:

• Corporate data theft
• Ransomware
• Insider attacks
• Website compromise
• Intellectual property theft

3: Crimes against government

Examples:

• Attacks against government websites
• Espionage

• Theft of government information
• Disruption of government services

4: Crimes against society

Examples:

• Large scale online fraud
• Distribution of illegal content
• Coordinated misinformation campaigns
• Attacks affecting public services

LESSON 3: TYPES OF CYBERCRIME

1 Unauthorised access

Unauthorised access occurs when a person accesses a computer system, account or network without lawful permission.

Example

A student obtains another student's password and logs into the student's university portal.

Discussion

Even if the student does not steal anything, unauthorised access may still be unlawful.

2 Phishing

Phishing involves attempts to deceive individuals into revealing sensitive information.

Typical targets include:

• Passwords
• Banking information
• Login credentials
• Personal information

Practical example

A student receives:

"Your student portal account requires immediate verification."

The message contains a fraudulent link.

The student enters a username and password.

The criminal receives the information.

Prevention

Students should:

• Verify the sender
• Avoid suspicious links
• Check website addresses carefully
• Use multifactor authentication
• Report suspicious messages

3 Identity theft

Identity theft involves using another person's identifying information without lawful authority.

Information may include:

• Name
• Identification details
• Account information
• Login credentials
• Personal documents

4 Online financial fraud

This includes schemes designed to unlawfully obtain money through electronic communication or digital platforms.

Examples include:

• Fake investment schemes
• Payment fraud
• Business email compromise
• Impersonation scams
• Fraudulent online transactions

5 Malware

Malware means malicious software.

Examples include:

• Viruses
• Worms
• Trojans
• Ransomware
• Spyware
• Some forms of malicious scripts

Ransomware

Ransomware can prevent an organisation from accessing its files and demand payment.

Classroom example

Suppose a university department suddenly discovers that important administrative files cannot be opened.

Students should ask:

1. What happened?
2. What evidence should be preserved?
3. Should the affected computer immediately be reformatted?
4. Who should be notified?
5. What backups are available?

6 Denial of Service

A denial of service attack attempts to make a service unavailable to legitimate users.

A distributed denial of service attack uses multiple systems or sources.

Example

A university website becomes unavailable during registration because it is receiving an abnormal volume of requests.

7 Cyberstalking and online harassment

Technology can be misused to repeatedly intimidate, threaten or harass another person.

Universities should have policies for reporting such incidents.

8 Intellectual property crime

Digital technology makes it easy to copy and distribute:

- Software
- Books
- Videos
- Research materials
- Music
- Databases

Students should understand that easy copying does not automatically make copying lawful.

9 Insider threats

An insider threat occurs when someone with legitimate access misuses that access.

Example:

An employee with authorised access to student records copies confidential information and gives it to an unauthorised person.

This demonstrates an important principle:

Authorised access does not mean unlimited authority.

LESSON 4: THREATS TO NATIONAL CRITICAL INFRASTRUCTURE

1 Meaning of critical infrastructure

Critical infrastructure refers to systems and services that are essential to the functioning of a country or society.

Examples can include:

- Electricity
- Telecommunications
- Banking and financial services
- Transportation
- Healthcare
- Government services
- Water systems
- Emergency services
- Major information and communication systems

2 Why critical infrastructure is attractive to attackers

An attack against a critical service may affect many people simultaneously.

For example:

If a hospital's information system becomes unavailable, patient care may be affected.

If a major financial system is disrupted, businesses and citizens may experience serious consequences.

3 Types of threats

Cyber espionage

Unauthorised collection of sensitive information.

Sabotage

Intentional disruption or destruction of systems.

Ransomware

Malicious software that disrupts access to data or systems.

Distributed denial of service

Attempts to overwhelm online services.

Supply chain compromise

An attacker compromises a supplier or third party to gain access to the final organisation.

CLASS ACTIVITY

Divide the class into groups.

Assign each group one sector:

1. Banking
2. Healthcare
3. Electricity
4. University
5. Telecommunications

Ask each group to identify:

• Three important systems
• Three possible cyber threats
• Possible consequences
• Five preventive measures

LESSON 5: CYBERCRIME PERPETRATORS AND THEIR STRATEGIES

1 Who commits cybercrime?

Cybercrime perpetrators can include:

• Individuals
• Organised criminal groups
• Disgruntled insiders
• Fraud networks
• Ideologically motivated groups
• State associated actors
• Opportunistic criminals

It is important not to assume that all cybercriminals have the same motivation.

2 Common motivations

Financial gain

The offender wants money.

Revenge

The offender wants to harm an organisation or individual.

Espionage

The objective is to obtain information.

Political or ideological objectives

The offender wants to promote a particular cause.

Recognition

Some offenders may seek reputation within an online criminal community.

3 Common social engineering strategies

Cybercrime frequently exploits human psychology.

Common techniques include:

Urgency

"You must act immediately."

Fear

"Your account will be closed."

Authority

"I am calling from the bank."

Curiosity

"See the attached document."

Reward

"You have won a prize."

PRACTICAL CLASSROOM EXERCISE

Give students this scenario:

A staff member receives an email apparently from the university administrator requesting confidential information urgently.

Ask:

1. What warning signs should the staff member look for?
2. How should the request be verified?
3. What should the staff member do if the message appears fraudulent?
4. What evidence should be preserved?